

# *ES120 COMSOL Project*

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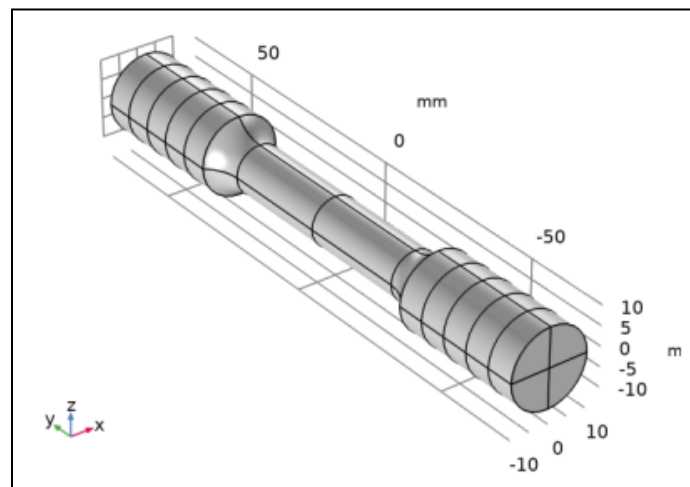
ES120: Introduction to Mechanics of Solids  
Spring 2026

Prof. M. Coughlin

## **Background:**

The purpose of this project was to prove two foundational principles of solid mechanics: 1) the linearized model of equations of elasticity and 2) Saint-Venant's Principle. Through designing and simulating experiments in COMSOL, we collected and analyzed the resulting data to draw quantitative conclusions for each principle.

For both Assumption I and II, we utilized the test specimen model provided for the project, which is a standardized model for deformation testing. Additionally, we generalized the material type to aluminum as it is used in construction applications for its weight and energy efficiency (Savin).



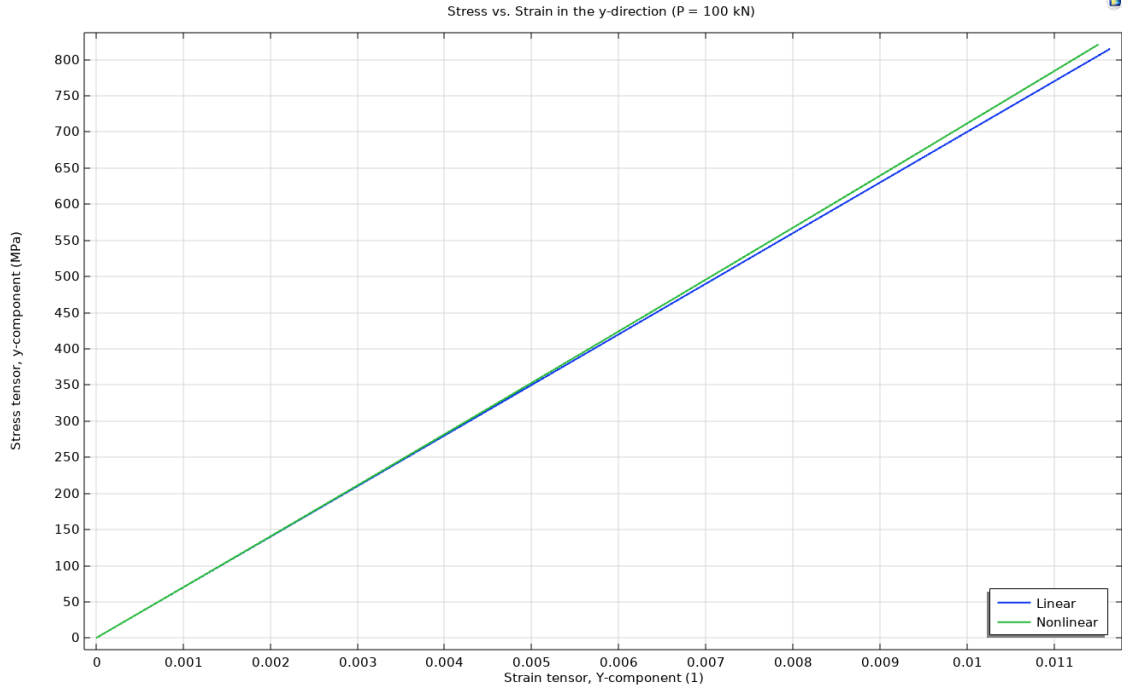
**Fig. 1:** Provided ASTM E8 round specimen model in COMSOL.

## **I. Assumption I**

The linearization of the equations of elasticity allows for a simplified analysis of the stress and strain of a material undergoing small deformations. When assuming elasticity, we can utilize the linear relationship between stress ( $\sigma$ ) and strain ( $\epsilon$ ) defined by Hooke's Law:  $\sigma = E\epsilon$ , where  $E$  denotes Young's modulus.

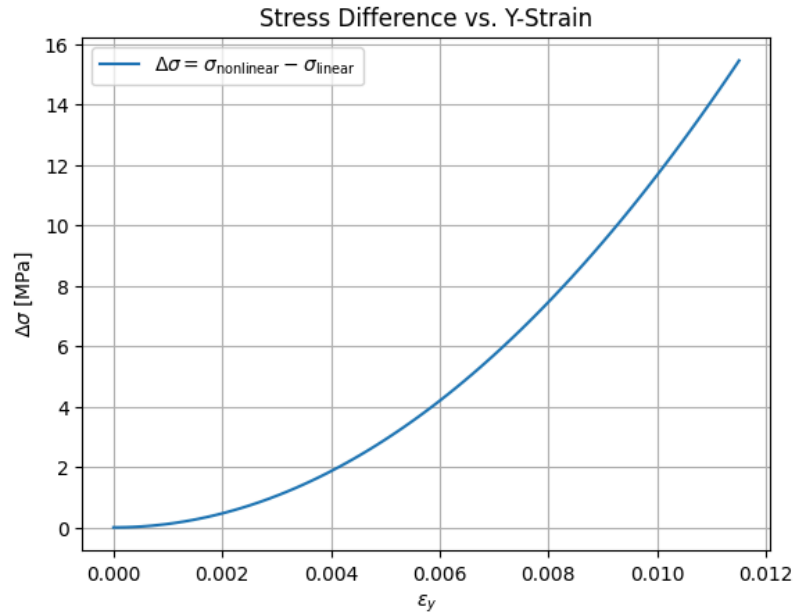
In order to confirm the validity of this linear approximation, we tested axial deformation on a test specimen in COMSOL to compare the effects of linear vs. nonlinear geometry on the resulting stress-strain curve. In particular, we wanted to quantitatively determine the point at which the linear and nonlinear models deviated and see if that point was within the working range (i.e., the elastic range) of the material.

After fixing one end of the test specimen and applying a boundary load to the other end, we ran a parametric sweep of the applied load from 0 N to 100 kN, and used a Cutpoint 3D at the midpoint of the model to extract stress-strain data in the y-direction.



**Fig. 2:** Stress-strain graph for a max applied load of 100 kN.

From the above graph, we observed that the linear and nonlinear models began to deviate at around  $(\epsilon_{sep}, \sigma_{sep}) = (0.005, 350 \text{ MPa})$ , with significant separation at  $\epsilon_y > 0.008$ . We further validated these findings through plotting the difference in the linear and nonlinear stress against the strain.



**Fig. 3:** Difference in nonlinear and linear stress vs. strain for a max load of 100 kN.

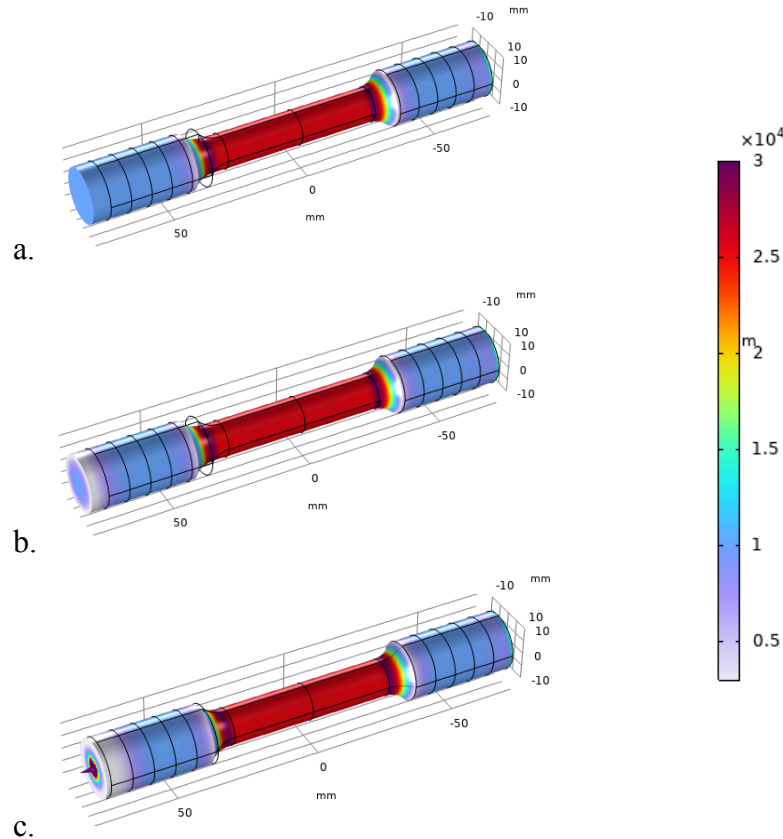
The yield strength ( $\sigma_Y$ ) of aluminum occurs at around 240-280 MPa (Modulus Metal), which is less than the separation stress of  $\sigma_{sep} = 350$  MPa at  $\epsilon_{sep} = 0.005$ , and significantly less than the noticeable deviation of  $\sigma_Y \approx 560$  MPa at  $\epsilon_Y = 0.008$ . Since  $\sigma_{sep} \gg \sigma_Y$ , we can conclude that the material will fail long before nonlinear effects dominate. Thus, the linear approximation of elasticity is reasonable in working stress-strain applications in the linear elastic range.

## II. Assumption II

St-Venant's principle states that statically equivalent loads will produce the same stress distribution past the immediate vicinity of the load, no matter the type of load distribution. There is not a quantitative explanation for what the immediate vicinity is defined as, and in this study, we sought to quantify the distance away from a load for which St-Venant's principle still holds.

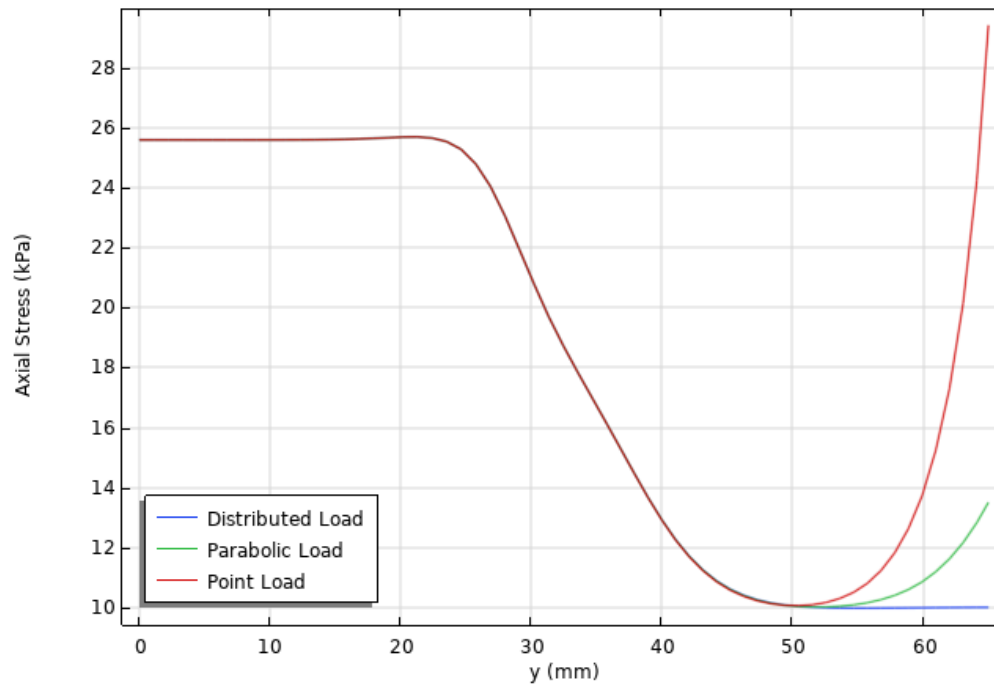
Using the same model and material from Assumption I, we tested three different axial loads on the face at  $y = 72.5$ mm: (a) a uniformly distributed load of 10 MPa, (b) a parabolic load of  $2 \cdot (10 \text{ MPa}) \cdot (1 - ((x^2 + z^2) / R^2))$ , and (c) a point load of  $(10 \text{ MPa})(\pi R^2)$  at the center of the face. These three load types are all statically equivalent to a force of  $(10 \text{ MPa})(\pi R^2) = 3141.59 \text{ N}$ . We ran studies for each of these loads with an extremely fine mesh to solve for the stress distribution throughout the entire volume (Fig. 4).

When the same colorbar scale is used, all the load types have a similar-looking stress distribution, with differences only appearing on and near the face where the load is applied, which matches with St. Venant's principle.



**Fig. 4:** Von Mises stress ( $\text{N/m}^2$ ) throughout volume with a statically equivalent (a) distributed load, (b) parabolic load, and (c) point force.

To quantify the difference in stress distribution of the different load types, we took a Cut Line 3D along the axis of the model from  $y = 0\text{mm}$  to  $y = 65\text{mm}$  and plotted the axial stress (Fig. 5). Since the load is applied at  $y = 72.5\text{mm}$ , this graph shows the stress at a location both close to the load and far from the load where the stress no longer shows any differences. Based on this graph, the point where the stresses converge appears to be at  $y = 50\text{mm}$ . This tells us that for this model geometry, material, and load value, St. Venant's principle is valid past a distance of  $22.5\text{mm}$  from the load application. Further tests could be performed to determine how this metric changes for different geometries and load amounts.



**Fig. 5:** Stress profile along center axis for different load types.

## Works Cited

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